GM/T 0004

SM3 Cryptographic Hash Algorithm

Cryptography Standardization
Technical Committee of China

Issued on 2012-03-21 Translated on 2024-10-30

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SM3 Cryptographic Hash Algorithm

1 Scope

This document specifies the SM3 cryptographic hash algorithm, including the details of the computation process and the computation examples.

This document applies to cryptographic applications using cryptographic hash algorithm, such as the digital signatures, the generation of the message authentication codes and the pseudorandom numbers (bits).

2 Terms and definitions

The following terms and definitions apply to this document.

2.1
bit string

A sequence of binary digits with values of 0 or 1.

2.2
big-endian

A data representation format in computer memory where the most significant bit (leftmost) is stored at the lowest memory address, and the least significant bit (rightmost) is stored at the highest memory address.

2.3
message

Any bit string of finite length.

Note 1: In this document, a message is used as the input data of the hash algorithm.

2.4
hash value

The output of a cryptographic hash algorithm, also called the digest.

Note 2: The length of the hash value in this document is 256 bits.

2.5

word

A bit string of length 32.

3 Symbols

The following symbols apply to this document.

$ABCDEFGH$: the concatenation of eight 32-bit registers or their values.

$B^{(i)}$: the i-th message block.

CF: a compression function.

FF_j: a Boolean function with an expression that varies depending on j.

GG_j: a Boolean function with an expression that varies depending on j.

IV: the initial value, used to initialize the register of the compression function.

P_0: the permutation involved in the compression function.

P_1: the permutation involved in the message expansion.

T_j: a set of constants, that vary depending on j.

m: a message.

m': a message after padding.

n: the number of blocks in a padded message.

mod: the modular operation.

$\wedge$: 32-bit bitwise AND operation.

$\vee$: 32-bit bitwise OR operation.

$\oplus$: 32-bit bitwise XOR operation.

$\neg$: 32-bit bitwise NOT operation.

$+$: addition modulo 2^{32} operation.

$\lll k$: the 32-bit left circular shift by k bits.

$\leftarrow$: the left assignment operation.

4 Constants and functions

4.1 Initial Value

IV = `7380166f 4914b2b9 172442d7 da8a0600 a96f30bc 163138aa e38dee4d b0fb0e4e`

4.2 Constants

$$T_j = \begin{cases} 79cc4519 & 0 \le j \le 15, \\ 7a879d8a & 16 \le j \le 63. \end{cases}$$

4.3 Boolean Functions

$$FF_j(X,Y,Z) = \begin{cases} X \oplus Y \oplus Z & 0 \le j \le 15, \\ (X \wedge Y) \vee (X \wedge Z) \vee (Y \wedge Z) & 16 \le j \le 63. \end{cases}$$

$$GG_j(X,Y,Z) = \begin{cases} X \oplus Y \oplus Z & 0 \le j \le 15, \\ (X \wedge Y) \vee ((\neg X \wedge Z) & 16 \le j \le 63. \end{cases}$$

where X, Y, Z are 32-bit words.

4.4 Permutations

$P_0(X) = X \oplus (X \lll 9) \oplus (X \lll 17)$,

$P_1(X) = X \oplus (X \lll 15) \oplus (X \lll 23)$.

where X is a 32-bit word.

5 Algorithm description

5.1 Overview

For a message m of length l (where $l < 2^{64}$) in bits, the SM3 hash algorithm generates a 256-bit hash value through the processes of padding and iterative compression. See Annex A for examples of operation.

5.2 Padding

Assume the length of the message m is l bits. Firstly, append bit '1' to the end of the message. Then, append k bits of '0', such that k is the smallest non-negative integer satisfying $l + 1 + k \equiv 448 \pmod{512}$. Finally, append a 64-bit bit string, representing the binary form of the length l. The resulting padded message m' will have a bit length that is a multiple of 512.

EXAMPLE For the message 011000010110001001100011, with length l =24, the padded bit string is: $01100001\ \ 01100010\ \ 01100011\ \ 1\ \ \overbrace{00\cdots00}^{423\text{ bits}}\ \ \underbrace{\overbrace{00\cdots011000}^{64\text{ bits}}}_{\substack{\text{binary}\\ \text{form of } l}}$.

5.3 Iterative Compression

5.3.1 Iteration Procedure

The padded message m' is split into 512-bit blocks: $m' = B^{(0)}B^{(1)} \cdots B^{(n-1)}$, where $n = (l + k + 65)/512$. The iteration procedure for m' is as follows:

FOR $i = 0$ TO $n - 1$

$\quad V^{(i+1)} = CF\left(V^{(i)},\ B^{(i)}\right)$

ENDFOR

Here, CF is the compression function, $V^{(0)}$ is the 256-bit IV, and $B^{(i)}$ is the i-th block of the padded message. The result of the iterative procedure is $V^{(n)}$.

5.3.2 Message Expansion

The message block $B^{(i)}$ is expanded to 132 words $W_0,\ W_1, \cdots,\ W_{67},\ W_0^{'}, W_1^{'}, \cdots W_{63}^{'}$, for use in the compression function CF:

a. Split the message block $B^{(i)}$ into 16 words $W_0,\ W_1, \cdots, W_{15}$.

b. FOR $j = 16$ TO 67

$\quad W_j \leftarrow P_1\left(W_{j-16} \oplus W_{j-9} \oplus \left(W_{j-3} \lll 15\right)\right) \oplus \left(W_{j-13} \lll 7\right) \oplus W_{j-6}$

ENDFOR

c. FOR $j = 0$ TO 63

$\quad W_j^{'} = W_j \oplus W_{j+4}$

ENDFOR

5.3.3 Compression Function

Let $A,\ B,\ C,\ D,\ E,\ F,\ G,\ H$ be eight word registers, $SS1,\ SS2,\ TT1,\ TT2$ be four intermediate variables, and the compression function be $V^{(i+1)} = CF\left(V^{(i)},\ B^{(i)}\right)$ $(0 \leq i \leq n - 1)$.

The computation procedure is described as follows:

$ABCDEFGH \leftarrow V^{(i)}$

FOR $j = 0$ TO 63

$\quad SS1 \leftarrow \left((A \lll 12) + E + \left(T_j \lll (j \bmod 32)\right)\right) \lll 7$

$\quad SS2 \leftarrow SS1 \oplus (A \lll 12)$

$\quad TT1 \leftarrow FF_j(A, B, C) + D + SS2 + W_j^{'}$

$\quad TT2 \leftarrow GG_j(E, F, G) + H + SS1 + W_j$

$\quad D \leftarrow C$

$\quad C \leftarrow B \lll 9$

$B \leftarrow A$

$A \leftarrow TT1$

$H \leftarrow G$

$G \leftarrow F \lll 19$

$F \leftarrow E$

$E \leftarrow P_0(TT2)$

ENDFOR

$V^{(i+1)} \leftarrow ABCDEFGH \oplus V^{(i)}$

Here, the words are stored in big-endian format.

5.4 Hash Value

$ABCDEFGH \leftarrow V^{(n)}$

Output a 256-bit hash value: $y = ABCDEFGH$

Annex A

(informative)

Examples

A.1 Example 1

A.1.1 The input message is "abc", and its ASCII-coded version is:

```
616263
```

A.1.2 The message after the padding process is:

```
61626380 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000018
```

A.1.3 The message after the message expansion is:

$\boldsymbol{W_0 W_1 \cdots W_{67}}$:

```
61626380 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000018
9092e200 00000000 000c0606 719c70ed 00000000 8001801f 939f7da9 00000000
2c6fa1f9 adaaef14 00000000 0001801e 9a965f89 49710048 23ce86a1 b2d12f1b
e1dae338 f8061807 055d68be 86cfd481 1f447d83 d9023dbf 185898e0 e0061807
050df55c cde0104c a5b9c955 a7df0184 6e46cd08 e3babdf8 70caa422 0353af50
a92dbca1 5f33cfd2 e16f6e89 f70fe941 ca5462dc 85a90152 76af6296 c922bdb2
68378cf5 97585344 09008723 86faee74 2ab908b0 4a64bc50 864e6e08 f07e6590
325c8f78 accb8011 e11db9dd b99c0545
```

$\boldsymbol{W_0^{'} W_1^{'} \cdots W_{63}^{'}}$:

```
61626380 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000018 9092e200 00000000 000c0606 719c70f5
9092e200 8001801f 93937baf 719c70ed 2c6fa1f9 2dab6f0b 939f7da9 0001801e
b6f9fe70 e4dbef5c 23ce86a1 b2d0af05 7b4cbcb1 b177184f 2693ee1f 341efb9a
fe9e9ebb 210425b8 1d05f05e 66c9cc86 1a4988df 14e22df3 bde151b5 47d91983
6b4b3854 2e5aadb4 d5736d77 a48caed4 c76b71a9 bc89722a 91a5caab f45c4611
6379de7d da9ace80 97c00c1f 3e2d54f3 a263ee29 12f15216 7fafe5b5 4fd853c6
428e8445 dd3cef14 8f4ee92b 76848be4 18e587c8 e6af3c41 6753d7d5 49e260d5
```

A.1.4 **The intermediate values during the iterative compression are:**

j	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
	7380166f	4914b2b9	172442d7	da8a0600	a96f30bc	163138aa	e38dee4d	b0fb0e4e
0	b9edc12b	7380166f	29657292	172442d7	b2ad29f4	a96f30bc	c550b189	e38dee4d
1	ea52428c	b9edc12b	002cdee7	29657292	ac353a23	b2ad29f4	85e54b79	c550b189
2	609f2850	ea52428c	db825773	002cdee7	d33ad5fb	ac353a23	4fa59569	85e54b79
3	35037e59	609f2850	a48519d4	db825773	b8204b5f	d33ad5fb	d11d61a9	4fa59569
4	1f995766	35037e59	3e50a0c1	a48519d4	8ad212ea	b8204b5f	afde99d6	d11d61a9
5	374a0ca7	1f995766	06fcb26a	3e50a0c1	acf0f639	8ad212ea	5afdc102	afde99d6
6	33130100	374a0ca7	32aecc3f	06fcb26a	3391ec8a	acf0f639	97545690	5afdc102
7	1022ac97	33130100	94194e6e	32aecc3f	367250a1	3391ec8a	b1cd6787	97545690
8	d47caf4c	1022ac97	26020066	94194e6e	6ad473a4	367250a1	64519c8f	b1cd6787
9	59c2744b	d47caf4c	45592e20	26020066	c6a3ceae	6ad473a4	8509b392	64519c8f
10	481ba2a0	59c2744b	f95e99a8	45592e20	02afb727	c6a3ceae	9d2356a3	8509b392
11	694a3d09	481ba2a0	84e896b3	f95e99a8	9dd1b58c	02afb727	7576351e	9d2356a3
12	89cbcd58	694a3d09	37454090	84e896b3	6370db62	9dd1b58c	b938157d	7576351e
13	24c95abc	89cbcd58	947a12d2	37454090	1a4a2554	6370db62	ac64ee8d	b938157d
14	7c529778	24c95abc	979ab113	947a12d2	3ee95933	1a4a2554	db131b86	ac64ee8d
15	34d1691e	7c529778	92b57849	979ab113	61f99646	3ee95933	2aa0d251	db131b86
16	796afab1	34d1691e	a52ef0f8	92b57849	067550f5	61f99646	c999f74a	2aa0d251
17	7d27cc0e	796afab1	a2d23c69	a52ef0f8	b3c8669b	067550f5	b2330fcc	c999f74a
18	d7820ad1	7d27cc0e	d5f562f2	a2d23c69	575c37d8	b3c8669b	87a833aa	b2330fcc
19	f84fd372	d7820ad1	4f981cfa	d5f562f2	a5dceaf1	575c37d8	34dd9e43	87a833aa
20	02c57896	f84fd372	0415a3af	4f981cfa	74576681	a5dceaf1	bec2bae1	34dd9e43
21	4d0c2fcd	02c57896	9fa6e5f0	0415a3af	576f1d09	74576681	578d2ee7	bec2bae1
22	eeeec41a	4d0c2fcd	8af12c05	9fa6e5f0	b5523911	576f1d09	340ba2bb	578d2ee7
23	f368da78	eeeec41a	185f9a9a	8af12c05	6a879032	b5523911	e84abb78	340ba2bb
24	15ce1286	f368da78	dd8835dd	185f9a9a	62063354	6a879032	c88daa91	e84abb78
25	c3fd31c2	15ce1286	d1b4f1e6	dd8835dd	4db58f43	62063354	8193543c	c88daa91
26	6243be5e	c3fd31c2	9c250c2b	d1b4f1e6	131152fe	4db58f43	9aa31031	8193543c
27	a549beaa	6243be5e	fa638587	9c250c2b	cf65e309	131152fe	7a1a6dac	9aa31031
28	e11eb847	a549beaa	877cbcc4	fa638587	e5b64e96	cf65e309	97f0988a	7a1a6dac
29	ff9bac9d	e11eb847	937d554a	877cbcc4	9811b46d	e5b64e96	184e7b2f	97f0988a
30	a5a4a2b3	ff9bac9d	3d708fc2	937d554a	e92df4ea	9811b46d	74b72db2	184e7b2f
31	89a13e59	a5a4a2b3	37593bff	3d708fc2	0a1ff572	e92df4ea	a36cc08d	74b72db2
32	3720bd4e	89a13e59	4945674b	37593bff	cf7d1683	0a1ff572	a757496f	a36cc08d
33	9ccd089c	3720bd4e	427cb313	4945674b	da8c835f	cf7d1683	ab9050ff	a757496f
34	c7a0744d	9ccd089c	417a9c6e	427cb313	0958ff1b	da8c835f	b41e7be8	ab9050ff
35	d955c3ed	c7a0744d	9a113939	417a9c6e	c533f0ff	0958ff1b	1afed464	b41e7be8
36	e142d72b	d955c3ed	40e89b8f	9a113939	d4509586	c533f0ff	f8d84ac7	1afed464

```
37 e7250598 e142d72b ab87dbb2 40e89b8f c7f93fd3 d4509586 87fe299f f8d84ac7
38 2f13c4ad e7250598 85ae57c2 ab87dbb2 1a6cabc9 c7f93fd3 ac36a284 87fe299f
39 19f363f9 2f13c4ad 4a0b31ce 85ae57c2 c302badb 1a6cabc9 fe9e3fc9 ac36a284
40 55e1dde2 19f363f9 27895a5e 4a0b31ce 459daccf c302badb 5e48d365 fe9e3fc9
41 d4f4efe3 55e1dde2 e6c7f233 27895a5e 5cfba85a 459daccf d6de1815 5e48d365
42 48dcbc62 d4f4efe3 c3bbc4ab e6c7f233 6f49c7bb 5cfba85a 667a2ced d6de1815
43 8237b8a0 48dcbc62 e9dfc7a9 c3bbc4ab d89d2711 6f49c7bb 42d2e7dd 667a2ced
44 d8685939 8237b8a0 b978c491 e9dfc7a9 8ee87df5 d89d2711 3ddb7a4e 42d2e7dd
45 d2090a86 d8685939 6f714104 b978c491 2e533625 8ee87df5 388ec4e9 3ddb7a4e
46 e51076b3 d2090a86 d0b273b0 6f714104 d9f89e61 2e533625 efac7743 388ec4e9
47 47c5be50 e51076b3 12150da4 d0b273b0 3567734e d9f89e61 b1297299 efac7743
48 abddbdc8 47c5be50 20ed67ca 12150da4 3dfcdd11 3567734e f30ecfc4 b1297299
49 bd708003 abddbdc8 8b7ca08f 20ed67ca 93494bc0 3dfcdd11 9a71ab3b f30ecfc4
50 15e2f5d3 bd708003 bb7b9157 8b7ca08f c3956c3f 93494bc0 e889efe6 9a71ab3b
51 13826486 15e2f5d3 e100077a bb7b9157 cd09a51c c3956c3f 5e049a4a e889efe6
52 4a00ed2f 13826486 c5eba62b e100077a 0741f675 cd09a51c 61fe1cab 5e049a4a
53 f4412e82 4a00ed2f 04c90c27 c5eba62b 7429807c 0741f675 28e6684d 61fe1cab
54 549db4b7 f4412e82 01da5e94 04c90c27 f6bc15ed 7429807c b3a83a0f 28e6684d
55 22a79585 549db4b7 825d05e8 01da5e94 9d4db19a f6bc15ed 03e3a14c b3a83a0f
56 30245b78 22a79585 3b696ea9 825d05e8 f6804c82 9d4db19a af6fb5e0 03e3a14c
57 6598314f 30245b78 4f2b0a45 3b696ea9 f522adb2 f6804c82 8cd4ea6d af6fb5e0
58 c3d629a9 6598314f 48b6f060 4f2b0a45 14fb0764 f522adb2 6417b402 8cd4ea6d
59 ddb0a26a c3d629a9 30629ecb 48b6f060 589f7d5c 14fb0764 6d97a915 6417b402
60 71034d71 ddb0a26a ac535387 30629ecb 14d5c7f6 589f7d5c 3b20a7d8 6d97a915
61 5e636b4b 71034d71 6144d5bb ac535387 09ccd95e 14d5c7f6 eae2c4fb 3b20a7d8
62 2bfa5f60 5e636b4b 069ae2e2 6144d5bb 4ac3cf08 09ccd95e 3fb0a6ae eae2c4fb
63 1547e69b 2bfa5f60 c6d696bc 069ae2e2 e808f43b 4ac3cf08 caf04e66 3fb0a6ae
```

A.1.5 The hash value is:

```
66c7f0f4 62eeedd9 d1f2d46b dc10e4e2 4167c487 5cf2f7a2 297da02b 8f4ba8e0
```

A.2 Example 2

A.2.1 A message of 512 bits:

```
61626364 61626364 61626364 61626364 61626364 61626364 61626364 61626364
61626364 61626364 61626364 61626364 61626364 61626364 61626364 61626364
```

A.2.2 The message after the padding process is:

```
61626364 61626364 61626364 61626364 61626364 61626364 61626364 61626364
```

```
61626364 61626364 61626364 61626364 61626364 61626364 61626364 61626364
80000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000200
```

A.2.3 The first message block process is:

A.2.3.1 The message after the message expansion is:

$\boldsymbol{W_0 W_1 \cdots W_{67}}$:

```
61626364 61626364 61626364 61626364 61626364 61626364 61626364 61626364
61626364 61626364 61626364 61626364 61626364 61626364 61626364 61626364
a121a024 a121a024 a121a024 6061e0e5 6061e0e5 6061e0e5 a002e345 a002e345
a002e345 49c969ed 49c969ed 49c969ed 85ae5679 a44ff619 a44ff619 694b6244
e8c8e0c4 e8c8e0c4 240e103e 346e603e 346e603e 9a517ab5 8a01aa25 8a01aa25
0607191c 25f8a37a d528936a 89fbd8ae 00606206 10501256 7cff7ef9 3c78b9f9
cc2b8a69 9f03f169 df45be20 9ec5bee1 0a212906 49ff72c0 46717241 67e09a19
6efaa333 2ebae676 3475c386 201dcff6 2f18fccf 2c5f2b5c a80b9f38 bc139f34
c47f18a7 a25ce71d 42743705 51baf619
```

$\boldsymbol{W_0^{'} W_1^{'} \cdots W_{63}^{'}}$:

```
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 c043c340 c043c340 c043c340 01038381
c14040c1 c14040c1 01234361 c06303a0 c06303a0 29a88908 e9cb8aa8 e9cb8aa8
25acb53c ed869ff4 ed869ff4 20820ba9 6d66b6bd 4c8716dd 8041e627 5d25027a
dca680fa 72999a71 ae0fba1b be6fca1b 32697922 bfa9d9cf 5f29394f 03fa728b
06677b1a 35a8b12c a9d7ed93 b5836157 cc4be86f 8f53e33f a3bac0d9 a2bd0718
c60aa36f d6fc83a9 9934cc61 f92524f8 64db8a35 674594b6 7204b1c7 47fd55ef
41e25ffc 02e5cd2a 9c7e5cbe 9c0e50c2 eb67e468 8e03cc41 ea7fa83d eda9692d
```

A.2.3.2 The intermediate values during iterative compression are:

j	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
	7380166f	4914b2b9	172442d7	da8a0600	a96f30bc	163138aa	e38dee4d	b0fb0e4e
0	588b5dab	7380166f	29657292	172442d7	b2e561d0	a96f30bc	c550b189	e38dee4d
1	b31cecd3	588b5dab	002cdee7	29657292	887cdf53	b2e561d0	85e54b79	c550b189
2	087b31df	b31cecd3	16bb56b1	002cdee7	5234344f	887cdf53	0e85972b	85e54b79
3	17448b12	087b31df	39d9a766	16bb56b1	16372ca6	5234344f	fa9c43e6	0e85972b
4	dca06de5	17448b12	f663be10	39d9a766	f7bc113c	16372ca6	a27a91a1	fa9c43e6
5	8eb847a3	dca06de5	8916242e	f663be10	9fe64fb1	f7bc113c	6530b1b9	a27a91a1
6	0e0f1218	8eb847a3	40dbcbb9	8916242e	57e5fc4e	9fe64fb1	89e7bde0	6530b1b9

7	ada83827	0e0f1218	708f471d	40dbcbb9	55eb8591	57e5fc4e	7d8cff32	89e7bde0
8	6e12c163	ada83827	1e24301c	708f471d	c26a14b8	55eb8591	e272bf2f	7d8cff32
9	f7578117	6e12c163	50704f5b	1e24301c	3433dd28	c26a14b8	2c8aaf5c	e272bf2f
10	bc497c66	f7578117	2582c6dc	50704f5b	4f85c749	3433dd28	a5c61350	2c8aaf5c
11	ecc59168	bc497c66	af022fee	2582c6dc	8ce5ee61	4f85c749	e941a19e	a5c61350
12	63723715	ecc59168	92f8cd78	af022fee	38e2aa27	8ce5ee61	3a4a7c2e	e941a19e
13	e57bfbf8	63723715	8b22d1d9	92f8cd78	542318e7	38e2aa27	730c672f	3a4a7c2e
14	8ba504b1	e57bfbf8	e46e2ac6	8b22d1d9	a8c73777	542318e7	5139c715	730c672f
15	b6a4be20	8ba504b1	f7f7f1ca	e46e2ac6	8ae4d7a0	a8c73777	c73aa118	5139c715
16	c0a0e3f7	b6a4be20	4a096317	f7f7f1ca	f671e12a	8ae4d7a0	bbbd4639	c73aa118
17	68ef7357	c0a0e3f7	497c416d	4a096317	673f9d46	f671e12a	bd045726	bbbd4639
18	4c6499d3	68ef7357	41c7ef81	497c416d	f01924a3	673f9d46	0957b38f	bd045726
19	9f532735	4c6499d3	dee6aed1	41c7ef81	71c6ef02	f01924a3	ea3339fc	0957b38f
20	231d84bd	9f532735	c933a698	dee6aed1	108149de	71c6ef02	251f80c9	ea3339fc
21	6a203212	231d84bd	a64e6b3e	c933a698	90c31af9	108149de	78138e37	251f80c9
22	175c3b57	6a203212	3b097a46	a64e6b3e	508f82d2	90c31af9	4ef0840a	78138e37
23	cdcbabd5	175c3b57	406424d4	3b097a46	b5a2f2fb	508f82d2	d7cc8618	4ef0840a
24	7dd941f8	cdcbabd5	b876ae2e	406424d4	a541cb9b	b5a2f2fb	1692847c	d7cc8618
25	eaf54f3e	7dd941f8	9757ab9b	b876ae2e	912d4e17	a541cb9b	97ddad17	1692847c
26	f7310a83	eaf54f3e	b283f0fb	9757ab9b	b43da5e9	912d4e17	5cdd2a0e	97ddad17
27	f8441d7e	f7310a83	ea9e7dd5	b283f0fb	cf194872	b43da5e9	70bc896a	5cdd2a0e
28	270dce67	f8441d7e	621507ee	ea9e7dd5	7564b6c0	cf194872	2f4da1ed	70bc896a
29	ac12a6c0	270dce67	883afdf0	621507ee	964015e3	7564b6c0	439678ca	2f4da1ed
30	1bd9e6e3	ac12a6c0	1b9cce4e	883afdf0	0fac4cad	964015e3	b603ab25	439678ca
31	32418d74	1bd9e6e3	254d8158	1b9cce4e	3f717698	0fac4cad	af1cb200	b603ab25
32	9c89b505	32418d74	b3cdc637	254d8158	38766abf	3f717698	65687d62	af1cb200
33	3c60352a	9c89b505	831ae864	b3cdc637	8aedd93b	38766abf	b4c1fb8b	65687d62
34	2a116c70	3c60352a	136a0b39	831ae864	476048d4	8aedd93b	55f9c3b3	b4c1fb8b
35	a0c7c66f	2a116c70	c06a5478	136a0b39	b47a7dc5	476048d4	c9dc576e	55f9c3b3
36	b7e58f33	a0c7c66f	22d8e054	c06a5478	3a3537a9	b47a7dc5	46a23b02	c9dc576e
37	79baf4ca	b7e58f33	8f8cdf41	22d8e054	9455b731	3a3537a9	ee2da3d3	46a23b02
38	ad5b0bcf	79baf4ca	cb1e676f	8f8cdf41	289d35e0	9455b731	bd49d1a9	ee2da3d3
39	a167bd76	ad5b0bcf	75e994f3	cb1e676f	da27276b	289d35e0	b98ca2ad	bd49d1a9
40	2ccc1878	a167bd76	b6179f5a	75e994f3	7eded43b	da27276b	af0144e9	b98ca2ad
41	610c6084	2ccc1878	cf7aed42	b6179f5a	9da32cab	7eded43b	3b5ed139	af0144e9
42	a40209fe	610c6084	9830f059	cf7aed42	7d483846	9da32cab	a1dbf6f6	3b5ed139
43	6fa376a2	a40209fe	18c108c2	9830f059	12a851cf	7d483846	655ced19	a1dbf6f6
44	53f9ffc5	6fa376a2	0413fd48	18c108c2	c3d3327b	12a851cf	c233ea41	655ced19
45	4f60bbd5	53f9ffc5	46ed44df	0413fd48	f3cae7e6	c3d3327b	8e789542	c233ea41
46	6e89a7fb	4f60bbd5	f3ff8aa7	46ed44df	17394ca0	f3cae7e6	93de1e99	8e789542
47	fef3cb16	6e89a7fb	c177aa9e	f3ff8aa7	4a9e594f	17394ca0	3f379e57	93de1e99

```
48 fa8e6731 fef3cb16 134ff6dd c177aa9e 7d9e1966 4a9e594f 6500b9ca 3f379e57
49 08a826c3 fa8e6731 e7962dfd 134ff6dd ebfa90cc 7d9e1966 ca7a54f2 6500b9ca
50 614c7627 08a826c3 1cce63f5 e7962dfd 969ecf53 ebfa90cc cb33ecf0 ca7a54f2
51 d776618d 614c7627 504d8611 1cce63f5 423489f6 969ecf53 86675fd4 cb33ecf0
52 ef958266 d776618d 98ec4ec2 504d8611 6ef4554d 423489f6 7a9cb4f6 86675fd4
53 04b44fd2 ef958266 ecc31bae 98ec4ec2 290032b5 6ef4554d 4fb211a4 7a9cb4f6
54 008d6012 04b44fd2 2b04cddf ecc31bae 50aa1faa 290032b5 aa6b77a2 4fb211a4
55 57859fec 008d6012 689fa409 2b04cddf c00cd655 50aa1faa 95a94801 aa6b77a2
56 c864420d 57859fec 1ac02401 689fa409 2fb3c502 c00cd655 fd528550 95a94801
57 e7423482 c864420d 0b3fd8af 1ac02401 aac3b183 2fb3c502 b2ae0066 fd528550
58 5c5be9dd e7423482 c8841b90 0b3fd8af 8b1ba117 aac3b183 28117d9e b2ae0066
59 ebd4948c 5c5be9dd 846905ce c8841b90 74a75fe1 8b1ba117 8c1d561d 28117d9e
60 05627b53 ebd4948c b7d3bab8 846905ce f58d98d8 74a75fe1 08bc58dd 8c1d561d
61 28aaec87 05627b53 a92919d7 b7d3bab8 cc6b5f2a f58d98d8 ff0ba53a 08bc58dd
62 0f92d652 28aaec87 c4f6a60a a92919d7 b8ab6d40 cc6b5f2a c6c7ac6c ff0ba53a
63 2ad0c8ee 0f92d652 55d90e51 c4f6a60a 69caa1b7 b8ab6d40 f956635a c6c7ac6c
```

A.2.4 The second message block process is:

A.2.4.1 The message after the message expansion is:

$W_0 W_1 \cdots W_{67}$:

```
80000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000200
80404000 00000000 01008080 10005000 00000000 002002a0 ac545c04 00000000
09582a39 a0003000 00000000 00200280 a4515804 20200040 51609838 30005701
a0002000 008200aa 6ad525d0 0a0e0216 b0f52042 fa7073b0 20000000 008200a8
7a542590 22a20044 d5d6ebd2 82005771 8a202240 b42826aa eaf84e59 4898eaf9
8207283d ee6775fa a3e0e0a0 8828488a 23b45a5d 628a22c4 8d6d0615 38300a7e
e96260e5 2b60c020 502ed531 9e878cb9 218c38f8 dcae3cb7 2a3e0e0a e9e0c461
8c3e3831 44aaa228 dc60a38b 518300f7
```

$W'_0 W'_1 \cdots W'_{63}$:

```
80000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000200 80404000 00000000 01008080 10005200
80404000 002002a0 ad54dc84 10005000 09582a39 a02032a0 ac545c04 00200280
ad09723d 80203040 51609838 30205581 04517804 20a200ea 3bb5bde8 3a0e5517
10f50042 faf2731a 4ad525d0 0a8c02be caa105d2 d8d273f4 f5d6ebd2 828257d9
f07407d0 968a26ee 3f2ea58b ca98bd88 08270a7d 5a4f5350 4918aef9 c0b0a273
a1b37260 8ced573e 2e8de6b5 b01842f4 cad63ab8 49eae2e4 dd43d324 a6b786c7
```

c8ee581d f7cefc97 7a10db3b 776748d8 adb200c9 98049e9f f65ead81 b863c496

A.2.4.2 The intermediate values during iterative compression are:

j	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
	5950de81	468664eb	42fd4c86	1e7ca00a	c0a5910b	ae9a55ea	1adb8d17	763ca222
0	1cc66027	5950de81	0cc9d68d	42fd4c86	24fe81a1	c0a5910b	af5574d2	1adb8d17
1	b7197324	1cc66027	a1bd02b2	0cc9d68d	61b7397a	24fe81a1	885e052c	af5574d2
2	b1aacb3f	b7197324	8cc04e39	a1bd02b2	4c7cbb59	61b7397a	0d0927f4	885e052c
3	920d5d4d	b1aacb3f	32e6496e	8cc04e39	c6c863a3	4c7cbb59	cbd30db9	0d0927f4
4	03162191	920d5d4d	55967f63	32e6496e	dbcb73dd	c6c863a3	daca63e5	cbd30db9
5	cbfddbb7	03162191	1aba9b24	55967f63	6a6eaafb	dbcb73dd	1d1e3643	daca63e5
6	67f45147	cbfddbb7	2c432206	1aba9b24	e0cc5b97	6a6eaafb	9eeede5b	1d1e3643
7	dfc06393	67f45147	fbb76f97	2c432206	9d84a8d5	e0cc5b97	57db5375	9eeede5b
8	777f980d	dfc06393	e8a28ecf	fbb76f97	89d0a059	9d84a8d5	dcbf0662	57db5375
9	502a9be2	777f980d	80c727bf	e8a28ecf	befc3eda	89d0a059	46acec25	dcbf0662
10	df0f77ed	502a9be2	ff301aee	80c727bf	c8b999f7	befc3eda	02cc4e85	46acec25
11	b8bc2801	df0f77ed	5537c4a0	ff301aee	3a05da38	c8b999f7	f6d5f7e1	02cc4e85
12	5b3baaa5	b8bc2801	1eefdbbe	5537c4a0	eebf718f	3a05da38	cfbe45cc	f6d5f7e1
13	0f7185e4	5b3baaa5	78500371	1eefdbbe	f3fbf969	eebf718f	d1c1d02e	cfbe45cc
14	141cb1e7	0f7185e4	77554ab6	78500371	5cc495db	f3fbf969	8c7f75fb	d1c1d02e
15	f185448a	141cb1e7	e30bc81e	77554ab6	32028d02	5cc495db	cb4f9fdf	8c7f75fb
16	a7374acd	f185448a	3963ce28	e30bc81e	3d03e81b	32028d02	aedae624	cb4f9fdf
17	aaca2dcb	a7374acd	0a8915e3	3963ce28	130bc932	3d03e81b	68119014	aedae624
18	3d2dfd31	aaca2dcb	6e959b4e	0a8915e3	07fff8f8	130bc932	40d9e81f	68119014
19	15bab3e6	3d2dfd31	945b9755	6e959b4e	85b2dd34	07fff8f8	4990985e	40d9e81f
20	f477625b	15bab3e6	5bfa627a	945b9755	d2b3c82b	85b2dd34	c7c03fff	4990985e
21	ecbfba29	f477625b	7567cc2b	5bfa627a	604bda38	d2b3c82b	e9a42d96	c7c03fff
22	b9f6943d	ecbfba29	eec4b7e8	7567cc2b	e996d68b	604bda38	415e959e	e9a42d96
23	c537ac67	b9f6943d	7f7453d9	eec4b7e8	7f6c2bc6	e996d68b	d1c3025e	415e959e
24	c59665b3	c537ac67	ed287b73	7f7453d9	1a89ef0d	7f6c2bc6	b45f4cb6	d1c3025e
25	50115e1f	c59665b3	6f58cf8a	ed287b73	3ddf2899	1a89ef0d	5e33fb61	b45f4cb6
26	44196085	50115e1f	2ccb678b	6f58cf8a	0abc22da	3ddf2899	7868d44f	5e33fb61
27	bde4e355	44196085	22bc3ea0	2ccb678b	da96412a	0abc22da	44c9eef9	7868d44f
28	ca176dca	bde4e355	32c10a88	22bc3ea0	b418ac1b	da96412a	16d055e1	44c9eef9
29	541e456e	ca176dca	c9c6ab7b	32c10a88	35cf8215	b418ac1b	0956d4b2	16d055e1
30	b6feeef7	541e456e	2edb9594	c9c6ab7b	d41f5fda	35cf8215	60dda0c5	0956d4b2
31	026e42f7	b6feeef7	3c8adca8	2edb9594	c9436b11	d41f5fda	10a9ae7c	60dda0c5
32	8fd27582	026e42f7	fdddef6d	3c8adca8	a48dc4c2	c9436b11	fed6a0fa	10a9ae7c
33	2527f8c6	8fd27582	dc85ee04	fdddef6d	b29dc9d4	a48dc4c2	588e4a1b	fed6a0fa
34	3218579f	2527f8c6	a4eb051f	dc85ee04	0da81ad7	b29dc9d4	2615246e	588e4a1b
35	35421cf3	3218579f	4ff18c4a	a4eb051f	644b37e4	0da81ad7	4ea594ee	2615246e

```
36 12cb048f 35421cf3 30af3e64 4ff18c4a 107cb2fb 644b37e4 d6b86d40 4ea594ee
37 c6716749 12cb048f 8439e66a 30af3e64 7903974d 107cb2fb bf232259 d6b86d40
38 66bf4600 c6716749 96091e25 8439e66a e5575380 7903974d 97d883e5 bf232259
39 046516a9 66bf4600 e2ce938c 96091e25 e23d4f18 e5575380 ba6bc81c 97d883e5
40 e14ab898 046516a9 7e8c00cd e2ce938c 6e25affe e23d4f18 9c072aba ba6bc81c
41 bc44d883 e14ab898 ca2d5208 7e8c00cd 4ef0cb38 6e25affe 78c711ea 9c072aba
42 e017c779 bc44d883 957131c2 ca2d5208 10132c10 4ef0cb38 7ff3712d 78c711ea
43 11154e38 e017c779 89b10778 957131c2 c1d401bd 10132c10 59c27786 7ff3712d
44 3ba43e10 11154e38 2f8ef3c0 89b10778 953c1e65 c1d401bd 60808099 59c27786
45 445e8d34 3ba43e10 2a9c7022 2f8ef3c0 94bcdd11 953c1e65 0dee0ea0 60808099
46 34d09ee0 445e8d34 487c2077 2a9c7022 1d0ea72c 94bcdd11 f32ca9e0 0dee0ea0
47 18c77c40 34d09ee0 bd1a6888 487c2077 a8ca98c6 1d0ea72c e88ca5e6 f32ca9e0
48 a2507cea 18c77c40 a13dc069 bd1a6888 9845362a a8ca98c6 3960e875 e88ca5e6
49 7e014176 a2507cea 8ef88031 a13dc069 2cb0c2f2 9845362a c6354654 3960e875
50 eb39074b 7e014176 a0f9d544 8ef88031 0df22b74 2cb0c2f2 b154c229 c6354654
51 f67597e1 eb39074b 0282ecfc a0f9d544 8d4f6b2f 0df22b74 17916586 b154c229
52 31e9309d f67597e1 720e97d6 0282ecfc eecf99be 8d4f6b2f 5ba06f91 17916586
53 c6329c3c 31e9309d eb2fc3ec 720e97d6 c672ad96 eecf99be 597c6a7b 5ba06f91
54 75cc3800 c6329c3c d2613a63 eb2fc3ec 8515c87f c672ad96 cdf7767c 597c6a7b
55 925156ad 75cc3800 6538798c d2613a63 150cbd57 8515c87f 6cb63395 cdf7767c
56 7d0de10b 925156ad 987000eb 6538798c 7ee47610 150cbd57 43fc28ae 6cb63395
57 2066f136 7d0de10b a2ad5b24 987000eb 7d7aadcc 7ee47610 eab8a865 43fc28ae
58 85b31359 2066f136 1bc216fa a2ad5b24 07b9cfd1 7d7aadcc b083f723 eab8a865
59 6cddcb93 85b31359 cde26c40 1bc216fa c43eb29c 07b9cfd1 6e63ebd5 b083f723
60 23eff97d 6cddcb93 6626b30b cde26c40 1ea21d46 c43eb29c 7e883dce 6e63ebd5
61 07bd4e82 23eff97d bb9726d9 6626b30b c8d6867c 1ea21d46 94e621f5 7e883dce
62 64f3dc4a 07bd4e82 dff2fa47 bb9726d9 96e4028f c8d6867c ea30f510 94e621f5
63 87ee4178 64f3dc4a 7a9d040f dff2fa47 af7ee1ee 96e4028f 33e646b4 ea30f510
```

A.2.4.3 The hash value is:

```
debe9ff9 2275b8a1 38604889 c18e5a4d 6fdb70e5 387e5765 293dcba3 9c0c5732
```

———————————————